

SUJAL SAMAR DAS

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EDUCATION

B.Tech in Computer Engineering (CGPA: 8.98/10.00), SPIT, Mumbai	2022 – 2026
Minor in Management , SPJIMR (SP Jain Institute of Management and Research)	2023 – Present

SKILLS

Languages	Python , C++ , JavaScript , SQL
Web Development	React.js , Node.js , Express.js , WebSockets (Socket.IO) , RESTful APIs , MongoDB
Others	Flask, Pandas , NumPy , Scikit-learn, NLP
Developer Tools	Git , Docker , Linux , Jupyter , Vercel, Render , JWT Authentication

EXPERIENCE

Founder & Lead, Optificient Coding Initiative	Dec 2024 – Jun 2025
Sardar Patel Institute of Technology	Mumbai, India

- Initiated and led a student-run competitive programming group, organizing 10+ contests and technical workshops which boosted campus-wide engagement by **40%**.
- Established the community's vision by onboarding an initial cohort of 20+ members and implementing a mentorship program that guided 6 students in securing tech internships.

Teaching Assistant, Data Structures Algorithms	Jul 2024 – Nov 2024
Sardar Patel Institute of Technology	Mumbai, India

- Mentored **80+ undergraduate students** in weekly labs, clarifying core DSA concepts which contributed to a **15% improvement** in average exam scores.

PROJECTS

Rubik's Cube Optimal Solver	2025
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- Implemented the **Iterative Deepening A* (IDA*)** algorithm in **C++**, leveraging disjoint pattern database heuristics from Korf's research to aggressively prune the search space.
- Designed an efficient C++ representation of the cube's state and transitions, achieving optimal solutions for 13-move scrambles in under **10 seconds**.

Real-Time Collaborative Whiteboard	2025
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- Developed a full-stack, real-time whiteboard with a **Node.js** backend, leveraging **WebSockets (Socket.IO)** to achieve sub-100ms latency for simultaneous multi-user collaboration.
- Engineered RESTful APIs for session management, secured with **JWT authentication**, and deployed the stateful backend on **Render** and the **React** frontend on **Vercel**.

Plant Health Monitoring using Deep Learning (Major Project)	2025–Present
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- Built a **YOLOv5 + DeepLabv3+** pipeline to detect 5+ crop diseases from drone images, achieving **92% accuracy** in diverse field conditions.
- Designed a real-time monitoring dashboard with **Flask** and **React**, providing alerts projected to reduce crop loss by **20%** and increase yield by **15%**.

AWARDS AND ACHIEVEMENT

- Competitive Programming:** LeetCode **2110 (top 1.2%)** globally, Codeforces **1794 (Top 50, Mumbai)**; solved 1500+ problems.
- Rankings:** **1st Place** at ACM Semi-Code (2024), **3rd Place** at IEEE Code Hunt (2024), Top 7 Finalist at Code Uncode SPIT.
- CMI Scholarship (2022):** Awarded as one of only 5 recipients nationwide from a pool of over 10,000 applicants for excellence in mathematics.
- Selected for **AlgoZenith DSA Masterclass**, an invite-only program for advanced algorithm design mentored by industry experts.